LESSON PLAN

ENGINE HEALTH MONITORING (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Nuts, miscellaneous nuts, washers, Safety methods for turn buckles and various types of turn buckles
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Nuts, miscellaneous nuts, washers, Safety methods for turn buckles and various types of turn buckles.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe the use of various standards, identification of bolts, nuts, safety methods and maintenance of cables.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of nuts and washers; Demonstration of turn buckle safety methods; Inspection of turn buckle types
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a nut in an aircraft assembly?
(b) Name two types of miscellaneous nuts used in aviation.
(c) Why are turn buckles preferred over standard bolts for cable tensioning?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Standards for nuts and bolts • Identification of miscellaneous nuts • Function and types of washers • Safety methods for turn buckles • Types of turn buckles	L & Ds	PPT, IB	15 min
(b)	• Detailed examination of turn buckle mechanisms • Safety inspection criteria for turn buckles • Maintenance procedures for turn buckles • Comparison of various turn buckle types • Application of safety methods in practice	L & Ds	PPT, PDF	20 min
(c)	• Review of nut and washer standards • Reiteration of turn buckle safety protocols • Summary of miscellaneous nut applications • Troubleshooting common turn buckle issues • Final verification of learning objectives	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the safety method for securing a turn buckle to prevent slippage.
(b) Differentiate between a standard nut and a miscellaneous nut in terms of usage.
(c) Describe the maintenance steps required for a turn buckle after installation.

12. Home Assignment and Follow Up	Prepare a brief report on the identification and specifications of swaged type terminals as covered in the next lesson plan (LP 04).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Cable identification, specifications, replacement, swaged type terminals splicing, swaged terminals splicing, swaged ball terminals, NICO press process and thimble eye.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min